DYNAPRIM TABLET

MAL19910033AZ

DESCRIPTION:
TABLET

Colour	:	White
Shape	:	Round, flat and scored
Marking	:	"ST"

CONTENT:
Each Tablet Contains:
Sulphamethoxazole..................... 400 mg
Trimethoprim ....80 mg

PHARMACODYNAMICS:
Sulphamethoxazole and Trimethoprim interfere with the bacterial synthesis of tetrahydrofolic acid, and essential stage in the production of thymidine, purines, and subsequently nucleic acids. Sulphamethoxazole, like the other sulphonamides, inhibits the synthesis of dihydrofolic acid from p-aminobenzoic acid. Trimethoprim inhibits the action of dihydrofolate reductase and prevents the synthesis of tetrahydrofolic acid from dihydrofolic acid; although this stage also occurs in man, Trimethoprim is much less active against the mammalian enzyme. Enhanced antibacterial activity has been reported when Sulphamethoxazole and Trimethoprim are used together in vitro although there is doubt as to whether sequential blockage of the bacterial synthetic pathway is responsible. The clinical relevance of this enhanced activity is uncertain especially in urinary-tract infections where inhibitory concentrations of Trimethoprim appear to be dominant.

PHARMACOKINETICS:
Trimethoprim is readily absorbed from the GIT and peak concentrations occur about 3 hours after a dose is taken. About 45% of Trimethoprim is bound to plasma proteins. Tissue concentrations are reported to be higher than serum concentrations, with particularly high concentrations occurring in the kidneys and lungs but concentrations in the cerebrospinal fluid are about one-half of those in the blood. The half-life is about 10 to 16 hours. About 40 to 50% of a dose is excreted unchanged in the urine within 24 hours, together with metabolites. Trimethoprim appears in breast milk.
Sulphamethoxazole is readily absorbed from the GIT and peak plasma concentrations are reached within 4 hours. Doses of 1 g twice daily should produce blood concentrations of unconjugated Sulphamethoxazole in excess of 50 mcg per ml. About 65% is bound to plasma albumin and the plasma half-life is about 10 hours. About 15% of Sulphamethoxazole in the blood is present as the acetyl derivative. Elimination in the urine is dependent on pH. About 25% of a single 2 g dose of Sulphamethoxazole has been reported to be excreted in the urine within 8 hours, about 60% being in the form of the acetyl derivative.
When Sulphamethoxazole and Trimethoprim are administered, plasma concentrations of Trimethoprim and Sulphamethoxazole are generally in the ratio of 1:20; in urine this ratio may vary from 1:1 to 1:5. About 50% of administered Trimethoprim and 50% of Sulphamethoxazole is excreted in the urine in 24 hours; a larger proportion of Sulphamethoxazole appears as inactive metabolite.

INDICATION:
Indicated in the treatment of:
- Acute exacerbations of chronic bronchitis in adults caused by H. influenzae or streptococcus pneumoniae.
- Enterocolitis caused by Shigella flexneri and S. sonnei.
- Acute otitis media in children caused by H. influenzae or S. pneumoniae.
- Pneumoniae caused by Pneumocystic carinii.
- Urinary tract infections caused by E. Coli, Klebsiella species, Enterobacter, P. vulgaris and Morganella morgani.

RECOMMENDED DOSE:
Adults and children 32 Kg body-weight and over:

Antibacterial	:	2 tablets every 12 hours.
Antiprotozoal	:	4 tablets every 6 hours.

Maximum of 8 tablets daily.

Infants above 2 months and children up to 32 Kg body-weight.

Antibacterial	:	20 mg Sulphamethoxazole and 4 mg Trimethoprim per Kg body-weight every 12 hours.
Antiprotozoal	:	25 mg Sulphamethoxazole and 5 mg Trimethoprim per Kg body-weight every 6 hours.

Therapy should be continued for at least 10 – 14 days in acute exacerbations of chronic bronchitis, 7 – 10 days in urinary tract infections, 5 days in shigellosis, 10 days in acute otitis media in children, 14 days in pneumocystis carinii pneumonia. The medication can also be given as a single dose for 1 – 2 days for lower urinary tract infections. Therapy continued for 14 days for upper urinary tract infections.

ROUTE OF ADMINISTRATION:
FOR ORAL USE ONLY

CONTRAINDICATIONS:
It is contraindicated in infants under 1 month of age because Sulphamethoxazole may cause kernicterus in neonates. It should not be given to patients with a history of hypersensitivity to Sulphamethoxazole or Trimethoprim. It should be avoided in patients with haematological disorders, megaloblastic anaemia and bone marrow changes or folic-acid deficiency such as patients receiving anticonvulsant drugs. It should also be avoided in patients receiving oral anticoagulants.

WARNINGS AND PRECAUTIONS:

Fatalities associated with the administration of Sulphonamides and Trimethoprim, either alone or in combination, have occurred due to severe reactions, including Stevens-Johnson Syndrome, toxic epidermal necrolysis and other reactions. The drug should be discontinued at the first appearance of skin rash or any sign of adverse reaction.

It should be given with caution to patients with impaired renal or hepatic functions, G6PD deficiency, folate deficiency, severe allergy or bronchial asthma. Adequate fluid intake must be maintained in order to prevent crystalluria and stone formation. Administration of alkalis may be necessary if very large doses are used.

Respiratory toxicity
Very rare, severe cases of respiratory toxicity, sometimes progressing to Acute Respiratory Distress Syndrome (ARDS), have been reported during co-trimoxazole treatment. The onset of pulmonary signs such as cough, fever and dyspnoea in association with radiological signs of pulmonary infiltrates, and deterioration in pulmonary function may be preliminary signs of ARDS. In such circumstances, co-trimoxazole should be discontinued and appropriate treatment given.

DRUG INTERACTIONS:
Avoid in patients receiving anticonvulsant drugs or oral anticoagulants.
It should be used with caution in patients receiving pyrimethamine or immunosuppresive therapy.
Exposure to the direct sunlight should be avoided as it facilitates development of sensitisation dermatitis. Blood count should be done frequently and if a significant reduction in the count of any formed blood element is noted, then treatment should be discontinued. Urinalysis and renal function tests should be performed especially for those patients with impaired renal function and dosage may have to be reduced. It can enhance the anticoagulant effect of warfarin. It should be given with caution in patients receiving methotrexate, resulting in increased free methotrexate concentrations. Use with caution in patients with a history of hypersensitivity to other sulphonamides and sulphonamide derivative drugs.

PREGNANCY AND LACTATION:
It should be avoided in pregnant or breast-feeding mothers.

SIDE EFFECTS:
Gastrointestinal disturbances, headache, dizziness, drowsiness, drug fever, skin rash, acidosis, hypothyroidism, CNS disturbances, tinnitus, peripheral neuritis, polyarteritis nodosa. Renal complications such as lumbar pain, dysuria, haematuria, oliguria, and anuria, may result from crystallisation of Sulphamethoxazole or the acetyl derivative. Blood disorders have occasionally occurred include agranulocytosis, aplastic anaemia, and thrombocytopenia, leucopenia and oesinophilla. Allergic reactions such as the Stevens-Johnson and Lyell's syndromes may occur rarely.

SYMPTOMS AND TREATMENT OF OVERDOSE:
The maximum tolerated dose in human is unknown. Long period of administration may cause bone marrow depression manifested as thrombocytopenia, leukopenia and/or megaloblastic anaemia. If signs of the bone marrow depression occur, the patient should be given leucovorin 3 to 6 mg intramuscularly daily for three days, or as required to restore normal haematopoiesis is only moderately effective in eliminating Sulphamethoxazole and Trimethoprim.

PACKING/PACK SIZE(s):
Blister packs of 10x10's and 100x10's.

JAUHI UBAT DARIPADA KANAK-KANAK
KEEP OUT OF REACH OF CHILDREN

MANUFACTURER/PRODUCT REGISTRATION HOLDER:
DYNAPHARM (M) SDN. BHD. 198001011897 (65683-V)
2497, Mk. 1, Lorong Perusahaan Baru 5,
Kawasan Perusahaan Perai 3,
13600 Perai, Pulau Pinang, MALAYSIA

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